Digital System Design

Final Exam(96/06/15)
Spring 2009

Total points: 105 points

1. Correct the **errors** of the following Verilog codes. (15%)

```
module CheerUp (AndYou, IamFine, VeryGood, NotBad, Sel);
  output [1:0] IamFine, AndYou;
  input [1:0] VeryGood, NotBad;
  reg [1:0] VeryGood, NotBad;
  input Sel;
  reg [1:0] IamFine;

  // continue in the right hand side.
```

```
AndYou = VeryGood | NotBad;

always @( Sel or VeryGood)
 IamFine[0] = Sel? VeryGood [0]: VeryGood [1];
 IamFine[1] = Sel? NotBad[0]: NotBad[1];

endmodule
```

2. Please state what kind of hardware is inferred? Draw its symbol. (for example: AND gate, OR gate....) (10%)

(a)
```
reg outsig;
always @(en or ina)
  if (en) outsig=ina;
```

(b)
```
reg outsig;
always @(en or ina)
  if (en) outsig=ina;
  else    outsig=1'bz;
```

3. Write the Verilog code of the following modules.

(a) A 16-bit 2-to-1 multiplexer. (10%)
```
module MUX21(ina, inb, sel, out);
...
endmodule
```

(b) A 16-bit positive-edge-triggered D flip-flop with asynchronous reset. (10%)
```
module DFF16 (Q,D,clk,rst);
...
endmodule
```

(c) A 16-bit logic shifter whose function is defined as the following table. Note that you can not use "<<" or ">>" operators. (10%)

LR	Output	Description
0	ShOut=C>>1	Logic shift to the right by 1 bit
1	ShOut=C<<1	Logic shift to the left by 1 bit

***module** Shifter(ShOut,C,LR);*

...

endmodule

(d) An 16-bit arithmetic logic unit (ALU) having the following functions. (15%)

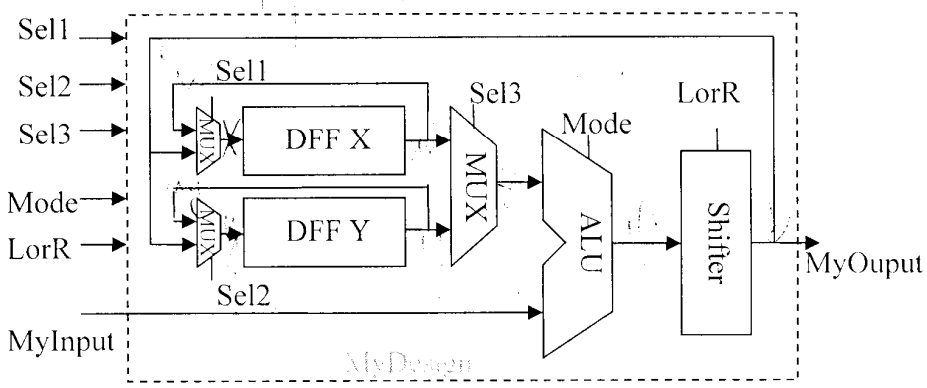
CTR	Output	Description
0	ALUOut=A+B	Addition
1	ALUOut=A-B	Subtraction
2	ALUOut=A+1	Increase by 1
3	ALUOut=A-1	Decrease by 1

***module** ALU(ALUOut, A, B, CTR);*

...

endmodule

4. Connect those instances in Question 2 as the following figures. Note that all the data paths are 16 bits. (20%)



5. Write test bench to test the above module called "MyDesign" as the following timing diagram. (15%)

